A full affirmative answer to the Grushin-plane no-axis-segment filling question

Candidate full-result packet for arXiv:1109.4641

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Status: candidate full result, likely valid. Every Lipschitz loop in the first Grushin plane whose image encloses no nondegenerate segment of the singular axis has a Lipschitz filling. Thus the result gives an affirmative answer to Question 8.7 of the source.

Abstract

We prove the remaining implication by unfolding the finite-length continuum traced by the loop. Whyburn's cyclic elements are its maximal cycle blocks. The no-enclosed-segment hypothesis forces every such block into one closed Grushin half-plane. The closed half-planes are CAT(0). Replacing each cycle block by a fresh copy of its half-plane, while retaining the dendritic connectors, produces a complete CAT(0) tree of spaces. The boundary loop lifts to that tree of spaces with controlled Lipschitz constant and the unfolding projects 1-Lipschitzly to the Grushin plane. Kirszbraun extension in the CAT(0) unfolding and projection complete the proof.

1 The source question and the result

The first Grushin plane G has underlying set $\mathbb{R}^2$, singular axis

$$Y = \{(0, y) : y \in \mathbb{R}\},$$

and, away from Y, length element

$$ds^2 = dx^2 + x^{-2}dy^2. \tag{1}$$

Equivalently, its orthonormal horizontal vector fields are ∂_x and $x\partial_y$. We write d_G for its Carnot–Caratheodory distance.

DeJarnette–Hajlasz–Lukyanenko–Tyson prove that every L-Lipschitz loop avoiding Y has an L-Lipschitz filling. They exhibit a bi-Lipschitz circle enclosing a nondegenerate segment of Y which has no Lipschitz filling, and ask [1, Question 8.7]:

> Riemannian metric, is complete and geodesic. The latter assertion follows from the explicit form (8.1) for the geodesics in G. Since $\{(x, y) \in G : x \geq \epsilon\}$ is also locally compact, it is proper and hence complete. The proof of Theorem 8.5 is finished. □
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> **Question 8.7.** Does every Lipschitz map $\varphi : \mathbb{S}^1 \to G$ whose image encloses no segment on the y-axis extend to a Lipschitz map of $\mathbb{B}^2$ into G?

Question 8.7 on PDF page 34 of the source.

We use the natural precise meaning of "encloses": a set $K \subset \mathbb{R}^2$ encloses a segment if a nondegenerate segment of Y is contained in a bounded component of $\mathbb{R}^2 \setminus K$. This is the meaning used by the source example.

2 Idea of the proof

The obstruction is not the number of times a loop meets the singular axis. It is whether a genuine cycle of its image uses both sides of that axis. The image is a finite-length Peano continuum, so Whyburn's cyclic-element theory splits it into maximal cycle blocks joined by dendritic connectors. A block that met both open half-planes would contain a cross-axis Jordan curve; the Jordan domain then forces the full image to enclose an axis interval. Therefore every block lies in one closed Grushin half-plane.

Each closed half-plane is CAT(0). We unfold the image by giving every cycle block its own copy of the appropriate half-plane and gluing the copies to the dendritic connectors only at their cut points. This is a tree gluing along convex singletons, so its completion is CAT(0). Boundary subarcs lift without length increase, and the unfolding projects 1-Lipschitzly back to G. Kirszbraun extension in the unfolded CAT(0) target followed by this projection gives the filling.

Theorem 1 (full answer to Question 8.7)**.** *Let $\varphi : \mathbb{S}^1 \to G$ be Lipschitz and put $K = \varphi(\mathbb{S}^1)$. If no nondegenerate segment of Y lies in a bounded component of $\mathbb{R}^2 \setminus K$, then φ extends to a Lipschitz map $F : \mathbb{B}^2 \to G$.*

If the unit circle is given its Euclidean chord metric and $\mathrm{Lip}(\varphi) = L$*, one may take* $\mathrm{Lip}(F) \le (\pi/2)L$.

The proof has three steps. First, each closed Grushin half-plane is CAT(0). Second, every genuine cycle block of K lies in one of those half-planes. Third, replacing cycle blocks by copies of their half-planes gives a CAT(0) unfolding to which the loop lifts.

3 Closed Grushin half-planes are CAT(0)

Set

$$G_+ = \{(x, y) \in G : x \ge 0\}, \qquad G_- = \{(x, y) \in G : x \le 0\}.$$

They carry the distance restricted from G. Folding x to $|x|$ preserves the length of every rectifiable Grushin curve whose endpoints are in G_+. Consequently the restricted distance on G_+ equals its intrinsic length distance. The corresponding statement holds for G_-.

Lemma 2 (half-plane CAT(0) lemma)**.** *The complete geodesic spaces G_+ and G_- are CAT(0).*

Proof. It is enough to treat G_+. For $\varepsilon > 0$, translate the smooth region

$$N_\varepsilon = \{(x, y) : x \ge \varepsilon\}$$

back to the fixed set G_+. Its pulled-back Riemannian metric is

$$g_\varepsilon = dx^2 + (x + \varepsilon)^{-2} dy^2, \tag{2}$$

and we denote its distance by d_ε.

The Gaussian curvature in the positive smooth Grushin region is $-2x^{-2} \le 0$. The region N_ε is complete and locally convex. For the latter assertion, the x-coordinate of a geodesic satisfies

$$x'' = -x^{-3}(y')^2 \le 0;$$

thus a geodesic segment with endpoints in $x \ge \varepsilon$ remains there. The metric Cartan–Hadamard theorem, or equivalently the standard convex boundary form of it, shows that (G_+, d_ε) is CAT(0).

As $\varepsilon \downarrow 0$, the metrics in (2) increase, so d_ε increases pointwise and is bounded above by d_G. In fact

$$d_\varepsilon(p, q) \uparrow d_G(p, q), \qquad p, q \in G_+. \tag{3}$$

Here is a direct compactness check. Take constant-speed almost minimizers $\gamma_\varepsilon = (x_\varepsilon, y_\varepsilon)$ for $d_\varepsilon(p,q)$. A uniform length bound controls x_ε and, on the resulting bounded x-range, also controls y_ε. After passing to a uniformly convergent subsequence, write

$$b_\varepsilon = \frac{y'_\varepsilon}{x_\varepsilon + \varepsilon}.$$

The controls $(x'_\varepsilon, b_\varepsilon)$ are bounded in the appropriate L^2 space. Weak compactness gives $y' = xb$ for the limit curve, including $y' = 0$ almost everywhere on $\{x = 0\}$, and lower semicontinuity gives

$$\mathrm{length}_G(\gamma) \le \liminf_{\varepsilon \downarrow 0} \mathrm{length}_{g_\varepsilon}(\gamma_\varepsilon).$$

This proves (3).

Pass now to the CAT(0) midpoint inequality. For fixed p, q, z, let m_ε be the d_ε-midpoint of p, q. The preceding compactness argument gives a subsequence converging to a d_G-midpoint m. On every relevant Euclidean compact set, (3) is uniform: the distance functions increase continuously to the continuous limit, so Dini's theorem applies. Hence

$$d_G(z,m)^2 \le \frac{1}{2} d_G(z,p)^2 + \frac{1}{2} d_G(z,q)^2 - \frac{1}{4} d_G(p,q)^2.$$

The space G_+ is complete (indeed it is closed in the proper Grushin plane), and the midpoint inequality characterizes complete CAT(0) spaces. Reflection proves the claim for G_-. □

Two immediate consequences will be used below. First, the identity map from either G_+ or G_- into G is an isometric embedding. Second, the Lang–Schroeder Kirszbraun theorem applies to maps from subsets of Euclidean space into either half-plane [2].

4 Finite-length continua and their cycle blocks

We record the classical continuum-theoretic input and its quantitative metric consequence. A *true cyclic element* of a Peano continuum X is a maximal nondegenerate subcontinuum C such that every two points of C lie on a Jordan curve contained in C.

We use the following standard facts from cyclic element theory [3, Chapter IV].

1. Every Jordan curve in X is contained in a true cyclic element.
2. Distinct true cyclic elements meet in at most one point, necessarily a cut point of X.
3. If C is a true cyclic element, the closure of each component of $X \setminus C$ meets C in at most one point.
4. After the true cyclic elements are regarded as blocks, the remaining connectors are dendritic and the block-incidence structure is a tree: there is a unique simple cyclic chain between any two elements.

For orientation, the point-attachment assertion in (3) is forced by maximality. If a component outside C attached at distinct points a, b, local connectedness would supply an outside arc from a to b. Combining it with a suitable arc in a Jordan curve of C would enlarge the cyclic element.

The next lemma packages exactly the infinite gluing statement needed here.

Lemma 3 (CAT(0) cyclic unfolding)**.** *Let M be a complete metric space and let $X \subset M$ be a compact connected set with $\mathcal{H}^1_M(X) < \infty$. For every true cyclic element $C \subset X$, suppose there are*

- *a complete CAT(0) space Z_C,*

- *a map* $i_C : C \to Z_C$*, and*
- *a* 1*-Lipschitz map* $p_C : Z_C \to M$

such that $p_C i_C$ *is the inclusion* $C \hookrightarrow M$*, and*

$$\mathrm{length}_{Z_C}(i_C \circ \alpha) \le \mathrm{length}_M(\alpha) \tag{4}$$

for every rectifiable path α *in* C*.*

Then there are a complete CAT(0) space E*, a* 1*-Lipschitz map* $P : E \to M$*, and a lift* $j : X \to E$ *with* Pj *equal to the inclusion, such that for every rectifiable path* α *in* X*,*

$$\mathrm{length}_E(j \circ \alpha) \le \mathrm{length}_M(\alpha). \tag{5}$$

Proof. A compact connected set of finite $\mathcal{H}^1$-measure is locally connected and rectifiably connected, hence is a Peano continuum. Apply its cyclic-element decomposition. Replace every true cyclic element C by a disjoint copy of Z_C. Retain each dendritic connector with its intrinsic arc-length metric, which is an $\mathbb{R}$-tree metric, and identify every attachment point with the corresponding point $i_C(q)$ in the adjacent block. Distinct blocks and connectors are therefore glued only along singletons. The tree property in (4) says that no cycle is introduced between the pieces.

For completeness, root the cyclic-element tree. Its nondegenerate block traces and nondegenerate connector components are countable: their interiors are disjoint, each has positive $\mathcal{H}^1$-measure, and their total measure is at most $\mathcal{H}^1(X)$. The remaining degenerate elements are limit points of this system. Take an increasing sequence of finite rooted subtrees that exhausts the nondegenerate pieces. The point-attachment property ensures that every piece newly exposed by a rooted subtree meets the preceding subtree in exactly its parent attachment point. If infinitely many pieces occur along one rooted chain, first include those whose *original traces in* X have diameter at least $1/n$, then let $n \to \infty$; there are only finitely many at each scale because $\mathcal{H}^1(X) < \infty$. This gives an increasing, dense union

$$E_1 \subset E_2 \subset \cdots$$

in which each new piece is attached to the old union at one point. Each finite stage is CAT(0) by Reshetnyak gluing along the closed convex set $\{q\}$, and its inclusion in the next stage is isometric and convex [4, Chapter II.11]. Every geodesic triangle in the union lies in some finite stage and satisfies the CAT(0) comparison inequality. Let E be its metric completion. Completions of CAT(0) spaces are CAT(0), so E is complete CAT(0). This is also the standard metric realization of the cyclic-element tree with the blocks C replaced by Z_C.

On Z_C define $P = p_C$. On every dendritic connector use its original map into $X \subset M$. These maps agree at attachment points. They are 1-Lipschitz on pieces, so the path-metric definition makes P short on the dense union; completeness of M extends it to a 1-Lipschitz map on E.

The canonical lift j sends a point of C to $i_C(C)$ and a point in the dendritic part to its corresponding tree point. Shared cut points were identified, so this is well defined and Pj is the inclusion. A rectifiable path in X lifts piece by piece. On block passages its length does not increase by (4); on dendritic passages the intrinsic arc-length is the original path length. First sum over finitely many passages and then exhaust the countably many passages. Monotone convergence of length gives (5). In particular, this estimate makes $j \circ \alpha$ continuous even at accumulating attachment points. □

Remark 4. *The finite-length hypothesis is doing two jobs: it makes* X *a Peano continuum, and it makes all nontrivial pieces countable with summable sizes. Thus the completion in Lemma 3 is not hiding a local-finiteness assumption.*

5 The no-segment condition makes every block one-sided

The metric induced on the singular axis has the snowflake scale

$$d_G((0,s),(0,t)) \asymp |s-t|^{1/2}. \tag{6}$$

Consequently every nondegenerate Euclidean interval in Y has infinite 1-dimensional Hausdorff measure for d_G.

Lemma 5 (one-sided cyclic elements)**.** *Let $K \subset G$ be compact, connected, and of finite $\mathcal{H}^1_G$-measure. Assume that no nondegenerate segment of Y is contained in a bounded component of $\mathbb{R}^2 \setminus K$. Then every true cyclic element of K is contained in G_+ or in G_-.*

Proof. Suppose a true cyclic element C contains a with positive x-coordinate and b with negative x-coordinate. By the defining property of C there is a Jordan curve $J \subset C$ through a and b. Let Ω be its bounded Jordan domain.

The set Ω meets Y. Otherwise connectedness would put Ω in one component of $\mathbb{R}^2 \setminus Y$, and then $J = \partial\Omega$ would lie in the corresponding closed half-plane, contrary to the choice of a, b. Since Ω is open, $\Omega \cap Y$ contains a nondegenerate interval I.

The finite-measure set K cannot contain an axis interval, by (6). Thus the compact set $K \cap I$ has empty relative interior in I, and $I \setminus K$ contains a nondegenerate open subinterval I_0. Because I_0 is connected, it lies in one component U of $\mathbb{R}^2 \setminus K$. The Jordan curve $J \subset K$ prevents U from leaving Ω, so U is bounded. Hence K encloses I_0, a contradiction. Therefore C cannot meet both open half-planes, proving the claim. □

6 Proof of the full theorem

Proof of Theorem 1. The image $K = \varphi(\mathbb{S}^1)$ is compact and connected, and

$$\mathcal{H}^1_G(K) \le \mathrm{length}_G(\varphi) < \infty.$$

By Lemma 5, every true cyclic element C of K is contained in one of G_+ and G_-. Choose one such half-plane and call it Z_C. Let $i_C : C \to Z_C$ and $p_C : Z_C \to G$ both be the coordinate identity. Lemma 2 says that Z_C is complete CAT(0). Folding shows that its intrinsic metric agrees with the distance restricted from G, so p_C is 1-Lipschitz and

$$\mathrm{length}_{Z_C}(i_C \circ \alpha) = \mathrm{length}_G(\alpha)$$

for every rectifiable path α in C.

Apply Lemma 3. We obtain a complete CAT(0) space E, a short map $P : E \to G$, and a lift $j : K \to E$ preserving the length upper bound on every rectifiable path. Put

$$\widetilde{\varphi} = j \circ \varphi : \mathbb{S}^1 \to E.$$

For $s, t \in \mathbb{S}^1$, let $A_{s,t}$ be the shorter circular arc between them. The length estimate in Lemma 3 gives

$$\begin{aligned} d_E(\widetilde{\varphi}(s), \widetilde{\varphi}(t)) &\le \mathrm{length}_E(\widetilde{\varphi}|_{A_{s,t}}) \\ &\le \mathrm{length}_G(\varphi|_{A_{s,t}}) \\ &\le L\ \mathrm{length}_{\mathbb{R}^2}(A_{s,t}) \le \frac{\pi}{2} L|s-t|. \end{aligned}$$

Thus $\widetilde{\varphi}$ is $(\pi/2)L$-Lipschitz on the unit circle with its Euclidean chord metric.

Euclidean space has curvature bounded below by 0, and E is complete CAT(0). The Lang–Schroeder Kirszbraun theorem therefore extends $\widetilde{\varphi}$ to a $(\pi/2)L$-Lipschitz map

$$\widetilde{F} : \mathbb{B}^2 \to E.$$

Finally set $F = P \circ \widetilde{F}$. Since P is 1-Lipschitz, F has the same Lipschitz bound, and on the boundary

$$F|_{\mathbb{S}^1} = Pj\varphi = \varphi.$$

This is the required filling. □

7 Stress tests and novelty check

The construction handles the configurations that defeat a direct one-contact gluing argument.

- A figure eight with one lobe on each side of Y becomes two CAT(0) half-planes glued at the common cut point.
- A theta block cannot mix the two sides: any two cross-side points in the same true cyclic element lie on a cross-axis Jordan curve, contradicting Lemma 5.
- Infinitely many alternating one-sided blocks cause no completeness problem. Finite subtrees are CAT(0), and their metric completion is CAT(0).
- A loop whose image has no Jordan curve at all lifts to an $\mathbb{R}$-tree, regardless of how often it meets the singular axis.

A bounded novelty search through 27 August 2026 covered the run registry and solution indexes; the exact wording of Question 8.7; searches combining "Grushin plane", "Lipschitz extension", "cyclic element", and "CAT(0)"; and the later Grushin embedding and quasiconformal papers returned by those searches. No later paper claiming this answer or this unfolding argument was found. Novelty confidence is moderate rather than definitive, because the proof combines classical ingredients in a short new way.

The packet is recommended for human review as a likely valid full answer. The highest-value audit is Lemma 3: compare its tree realization directly with Whyburn's cyclic-element decomposition, then verify that the finite-stage exhaustion includes every accumulating connector. The finite-length argument in Remark 4 is included precisely to expose that point.

References

[1] N. DeJarnette, P. Hajlasz, A. Lukyanenko, and J. T. Tyson, *On the lack of density of Lipschitz mappings in Sobolev spaces with Heisenberg target*, arXiv:1109.4641v3 (2014). https://arxiv.org/abs/1109.4641.

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[4] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Springer, 1999.